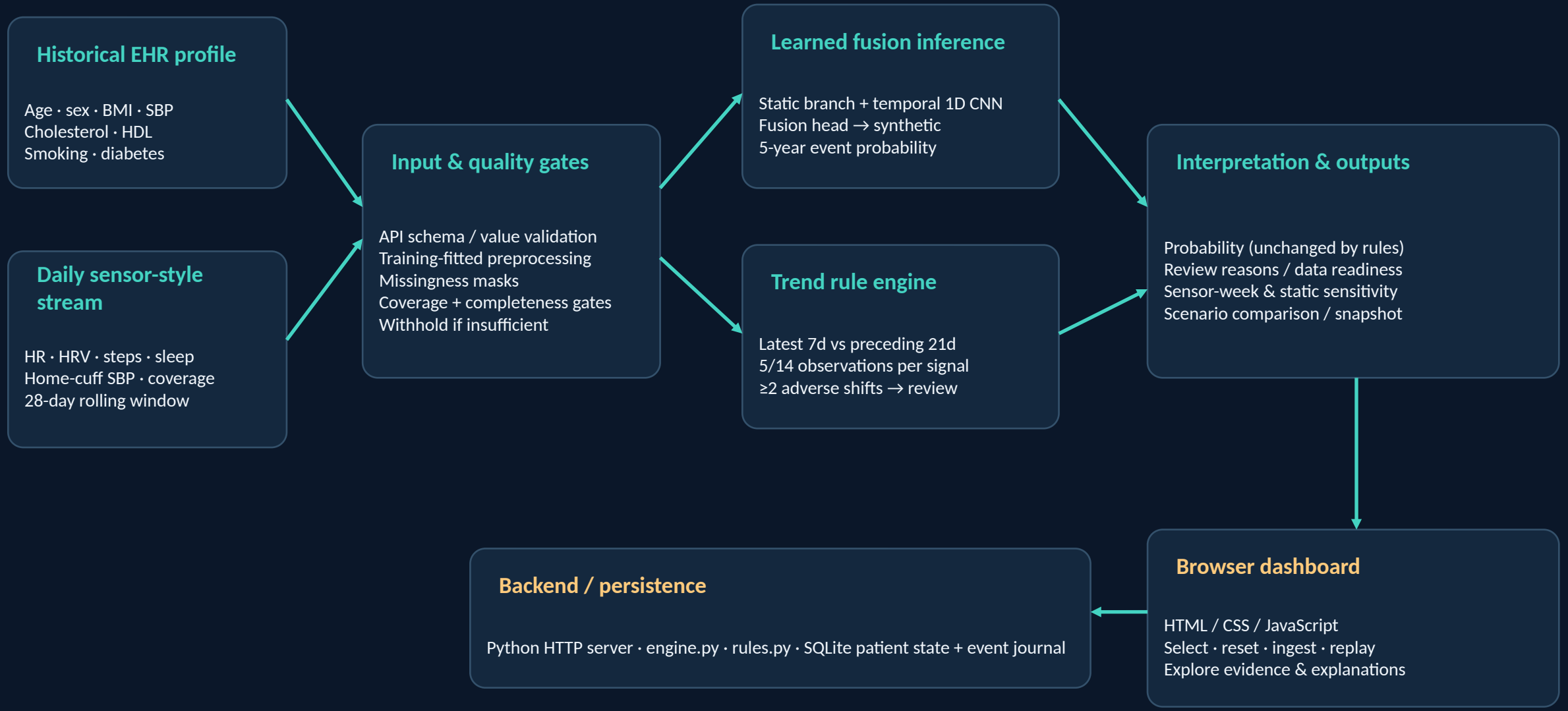
CARDIOTWIN | End-to-end architecture

Two linked synthetic data streams → validation → inference + review → persisted interactive dashboard



Important: all training and demo data are synthetic. The output is not a clinical diagnosis or imminent-event alert.

CARDIOTWIN | Dynamic update and decision trace

Operational detail: what happens when a fictional daily measurement is ingested

01 USER ACTION	Browser submits next-day fictional HR, HRV, steps, sleep, home BP and coverage.
02 VALIDATE	HTTP API validates payload, patient identity, acceptable fields and ranges.
03 PERSIST + ROLL	Append dated record; advance simulated date; retain most recent 28 days; journal update in SQLite.
04 CHECK READINESS	Check recent coverage, missingness and historical availability. Withhold output if inadequate.
05 MODEL INFERENCE	Apply saved train-fitted preprocessing; load selected checkpoint; fuse static + temporal branches.
06 RULE TRACE	Compare latest 7 vs previous 21 days; record per-signal reasons and multi-trend review flag.
07 EXPLAIN	Compute reference-replacement sensor/week and historical sensitivities; scenario does not mutate history.
08 RENDER	Return prediction or unavailable state, review reasons, before/after delta, trajectory and journal.

Evaluation is separate: held-out patient split → model metrics → hybrid policy comparison. Rule flags never change the CNN probability.